Project PDM

Introduction and Tips

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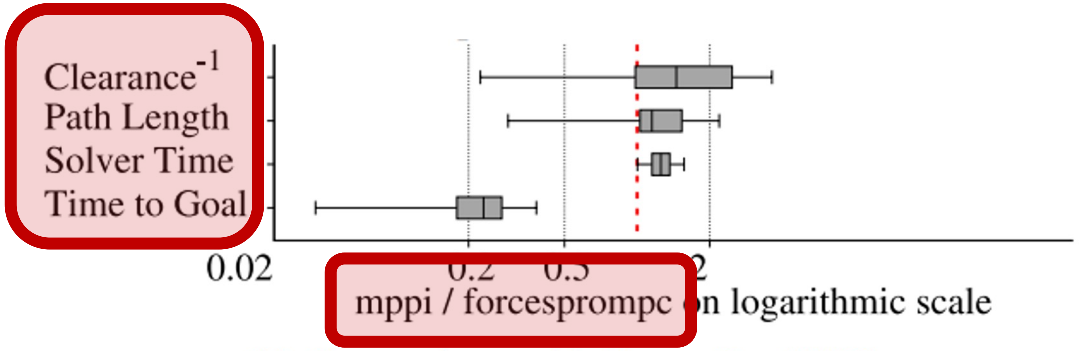
Project PDM

- Develop a full motion planning pipeline for a robot of your choice
  - Robot Morphology (car, manipulator, drone, etc.)
  - Motion Planner (RRT, MPC, etc.)
  - Environment (static, dynamic, cluttered, etc.)
- Validate performance
  - Set-up a test-bed with **randomized cases**
  - Evaluate your pipeline **quantitatively** and discuss/analyze results

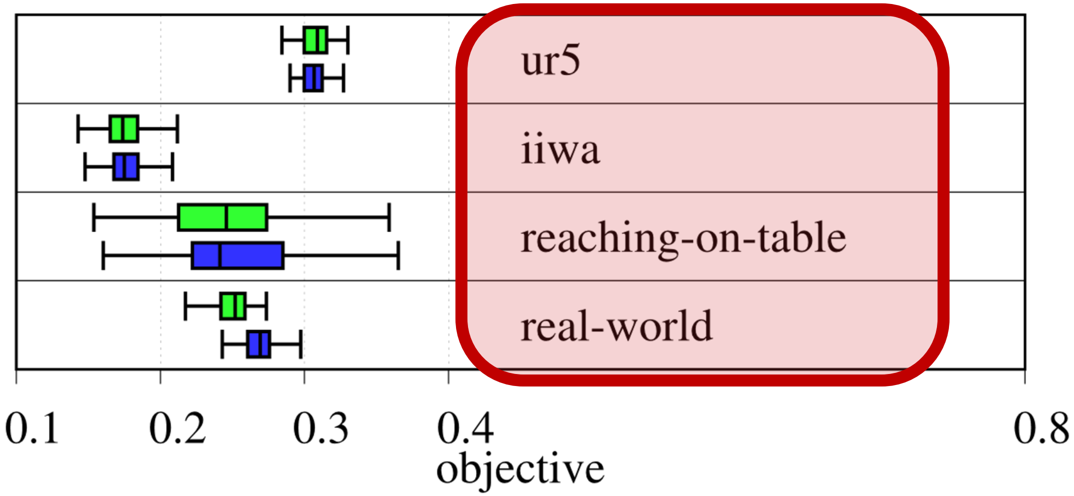
Results: Examples

TABLE II: Summary of comparison with [21]

Approach	Metric	Object A	Object B	Object C	Object D	Object E
Baseline [21]	Success [%]	93.5	90.9	93.9	91.6	**89.5**
	Time [s]	n/a	∼ 24	∼ 24	∼ 24	∼ 24
Ours (sim)	Success [%]	**100**	**93.3**	**96.7**	**100**	66.7
	Time [s]	**3.17** ±1.81	**3.52** ±1.84	**2.62** ±1.03	**2.46** ±0.47	**2.68** ±1.51



(b) Comparison with ForcesPro MPC



- **Metrics**
- **Comparison**
- **Scenarios**

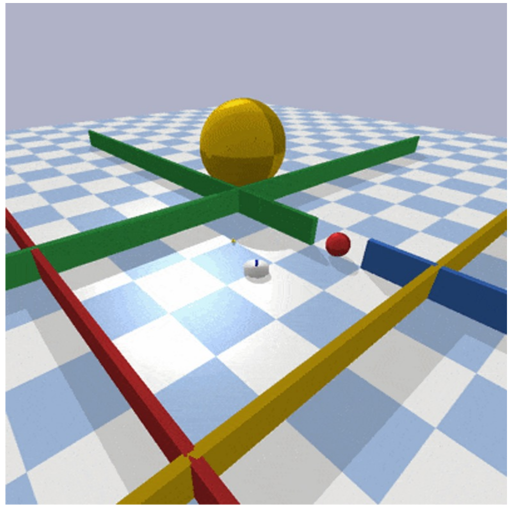
[1]: Sampling-based Model Predictive Control Leveraging Parallelizable Physics Simulations, Pezzato et al. 2023
[2]: Autotuning Symbolic Optimization Fabrics for Trajectory Generation, Spahn et al. 2023

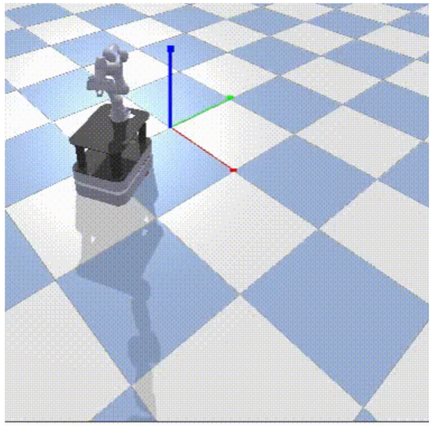
Process

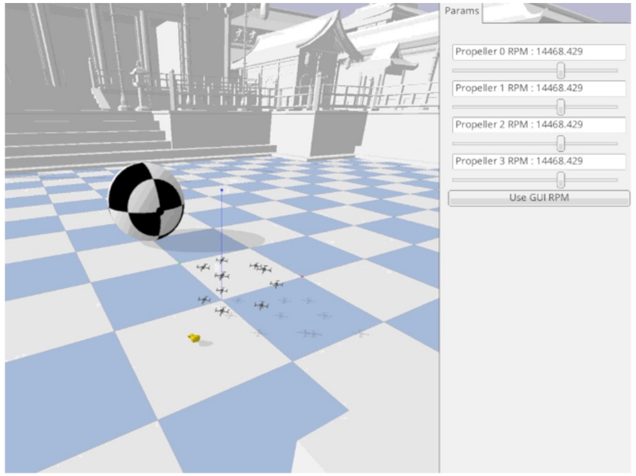
1. Choose your target scenario and described it carefully
2. Have a running case as quickly as possible (simplification of target)
   - Pick Simulation environment
   - Produce **quantitative results fast** for baseline
3. Analyse drawback of baseline
4. Develop your method for improving baseline

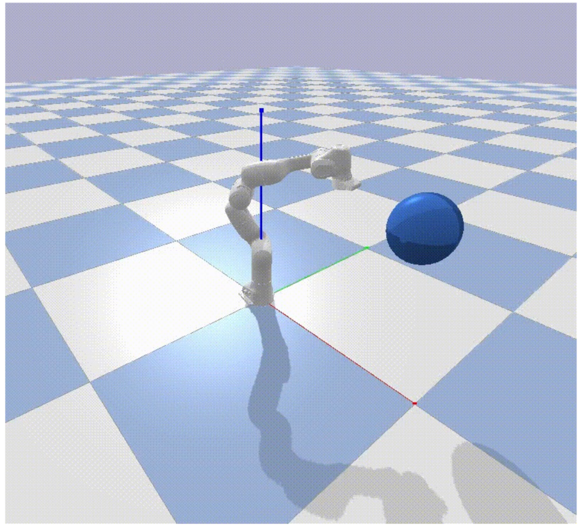
Simulation & Programming

- Recommend **UNIX** for robotics
- Recommend **python3** for this project
- Recommend **git** for version control and shared code
- Simulation Environments:
  - https://github.com/maxspahn/gym_envs_urdf
  - https://github.com/utiasDSL/gym-pybullet-drones
  - etc.









Getting Started

- Find a group
- Decide on problem formulation
- Installation issues
    - Help each other
    - Ask me if needed
- **ENJOY THE PROJECT!!!**